IIITB COMET FOUNDATION

Name:DARSI VAMSI
ID: COMETFWC071

Gate -PH 2011 46 Question

QUESTION

PH

The following Boolean expression
$Y = A \bullet \overline{B} \bullet \overline{C} \bullet \overline{D} + \overline{A} \bullet B \bullet \overline{C} \bullet D + \overline{A} \bullet \overline{B} \bullet \overline{C} \bullet D + \overline{A} \bullet \overline{B} \bullet C \bullet D + \overline{A} \bullet B \bullet C \bullet D + A \bullet \overline{B} \bullet \overline{C} \bullet D$
can be simplified to

(A) $\overline{A} \bullet \overline{B} \bullet C + A \bullet \overline{D}$
(B) $\overline{A} \bullet B \bullet \overline{C} + A \bullet \overline{D}$
(C) $A \bullet \overline{B} \bullet \overline{C} + \overline{A} \bullet D$
(D) $A \bullet \overline{B} \bullet C + \overline{A} \bullet D$

QUESTION ANALYSIS

Solution

can be simplified to

(A) $\overline{A} \cdot \overline{B} \cdot C + A \cdot \overline{D}$ **(B)** $\overline{A} \cdot B \cdot \overline{C} + A \cdot \overline{D}$

(C) $A \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot D$ **(D)** $A \cdot \overline{B} \cdot C + \overline{A} \cdot D$

Solution

Given,

$$Y = A\overline{BCD} + \overline{A}B\overline{C}D + \overline{ABC}D + \overline{A}\overline{BC}D + \overline{A}BCD + A\overline{BC}D$$

Step 1: Group the terms

Group the first and last terms:

$$A\overline{BCD} + A\overline{BC}D = A\overline{BC}(\overline{D} + D)$$

Since $\overline{D} + D = 1$,

$$A\overline{BC}(\overline{D} + D) = A\overline{BC}.$$

Now group the remaining four terms:

$$\overline{A}B\overline{C}D + \overline{ABC}D + \overline{A}\overline{BC}D + \overline{A}BCD$$
$$= \overline{A}D\left(B\overline{C} + \overline{BC} + \overline{B}C + BC\right).$$

Step 2: Simplify inside the brackets

$$
\begin{aligned}
& B\overline{C} + \overline{BC} + \overline{B}C + BC \\
&= \overline{C}(B + \overline{B}) + C(B + \overline{B}) \\
&= \overline{C}(1) + C(1) \\
&= \overline{C} + C \\
&= 1.
\end{aligned}
$$

Therefore,

$$\overline{A}D(1) = \overline{A}D.$$

Step 3: Final expression

Hence,

$$\boxed{Y = A\overline{BC} + \overline{A}D}$$

This matches **Option (C).**

$$\boxed{\text{Correct Answer: (C)}}$$

Hardware Components Required

S.No.	Component
1	ESP32-WROOM Development Board
2	Personal Computer (Ubuntu OS)
3	USB Cable
4	Breadboard
5	7447 BCD-to-7-Segment Decoder IC
6	Common Anode 7-Segment Display
7	220 Ω Resistors (7 Nos.)
8	Jumper Wires

Table 1: Hardware and Software Components Required

Hardware Connections

Execution Procedure

1. Create a new PlatformIO project for the ESP32-WROOM board.

```
/root/venv/bin/pio project init --board esp32dev
```

S.No.	Connection	Description
1	ESP32 GPIO16	7447 Input A
2	ESP32 GPIO17	7447 Input B
3	ESP32 GPIO18	7447 Input C
4	ESP32 GPIO19	7447 Input D
5	ESP32 GND	7447 GND
6	ESP32 5V (VIN)	7447 VCC
7	7447 Outputs (a–g)	7-Segment Display Segments (a–g)
8	Common Anode of 7-Segment	+5 V Supply
9	Segment Lines	Through 220 Ω Resistors
10	ESP32 USB Port	Ubuntu PC (Code Upload)

Table 2: Hardware Connections for ESP32-WROOM with 7447 and Common Anode 7-Segment Display

2. Place the source code (`main.cpp`) inside the `src` directory of the project.

3. Compile the source code and generate the binary files using:

   ```
   /root/venv/bin/pio run
   ```

4. After successful compilation, the following binary files are generated in the build directory:

   `.pio/build/esp32dev/`

   - `bootloader.bin`
   - `partitions.bin`
   - `firmware.bin`

5. Copy the generated binary files to the desired directory on the Ubuntu PC.

6. Connect the ESP32-WROOM board to the Ubuntu PC using a USB cable.

7. Verify that the ESP32 board is detected by the operating system.

   ```
   ls -l /dev/ttyUSB*
   ```

8. Verify the ESP32 chip.

   ```
   esptool.py --chip esp32 --port /dev/ttyUSB0 chip_id
   ```

9. Flash the generated binary files into the ESP32.

```
esptool.py --chip esp32 \
  --port /dev/ttyUSB0 \
  --baud 460800 \
  write_flash \
  0x1000 bootloader.bin \
  0x8000 partitions.bin \
  0x10000 firmware.bin
```

10. Wait until the flashing process completes successfully.
11. Press the **EN (Reset)** button on the ESP32 board if required.
12. Observe the LED connected to the output GPIO pin to verify the implemented Boolean expression.

Results and Observations

The simplified Boolean expression,

$$Y = A\overline{B}\,\overline{C} + \overline{A}D,$$

was successfully implemented on the ESP32-WROOM development board. The output of the Boolean function was converted into BCD format and applied to the 7447 BCD-to-7-segment decoder IC. The decoder drove a common anode 7-segment display to indicate the output of the Boolean expression. The generated binary files were successfully flashed onto the ESP32, and the experimental output matched the expected truth table for all possible input combinations.

Truth Table

Experimental Output

Figure 1: 7-Segment Display showing digit '0' corresponding to Logic LOW ($Y = 0$).

Figure 2: 7-Segment Display showing digit '1' corresponding to Logic HIGH ($Y = 1$).

Observation

Figure 3 shows the 7-segment display indicating the digit '0' when the output of the Boolean expression is logic LOW ($Y = 0$). Figure 4 shows the 7-segment display indicating the digit '1' when the output is logic HIGH ($Y = 1$). The observed outputs were in agreement with the theoretical truth table.

A	B	C	D	Y	7-Segment Display
0	0	0	0	0	0
0	0	0	1	1	1
0	0	1	0	0	0
0	0	1	1	1	1
0	1	0	0	0	0
0	1	0	1	1	1
0	1	1	0	0	0
0	1	1	1	1	1
1	0	0	0	1	1
1	0	0	1	1	1
1	0	1	0	0	0
1	0	1	1	0	0
1	1	0	0	0	0
1	1	0	1	0	0
1	1	1	0	0	0
1	1	1	1	0	0

Table 3: Truth Table of the Implemented Boolean Expression

Experimental Output

Result

The Boolean expression was successfully implemented and verified using the ESP32-WROOM development board, 7447 BCD-to-7-segment decoder IC, and common anode 7-segment display. The experimental results matched the expected outputs for all input combinations.

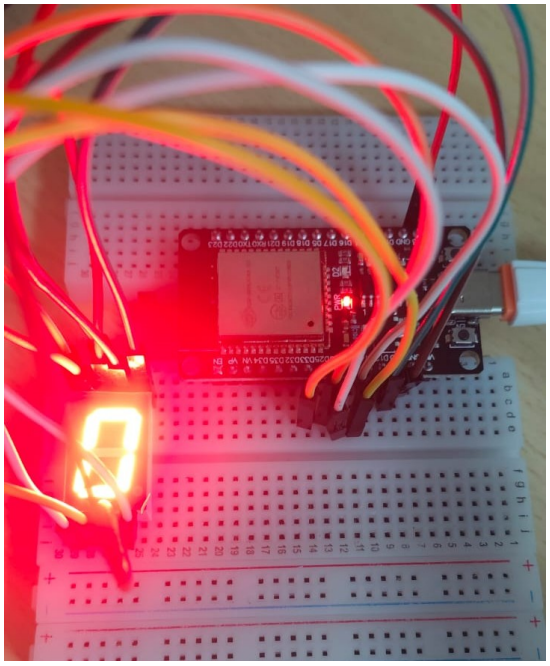


Figure 3: 7-Segment Display showing digit '0' corresponding to Logic LOW ($Y = 0$).

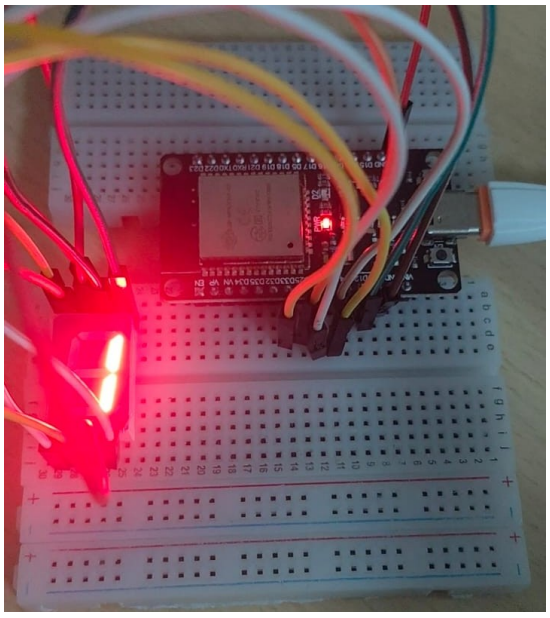


Figure 4: 7-Segment Display showing digit '1' corresponding to Logic HIGH ($Y = 1$).